

Term Design Project
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Introduction

Brushless DC (BLDC) motor drives operation requires control of the internal workings of the motor, this includes stator windings current and the trigger of the location of the rotor position. The example provided in class observed four Hall sensors place 90 degrees apart to achieve this, each energizing a directly opposite coil individually. Having a coil energizing sequence of A+, B+, A-, B-, repeat. The goal of this design project is to minimize the amount of Hall sensors and permanent magnets required to produce the same outcome.

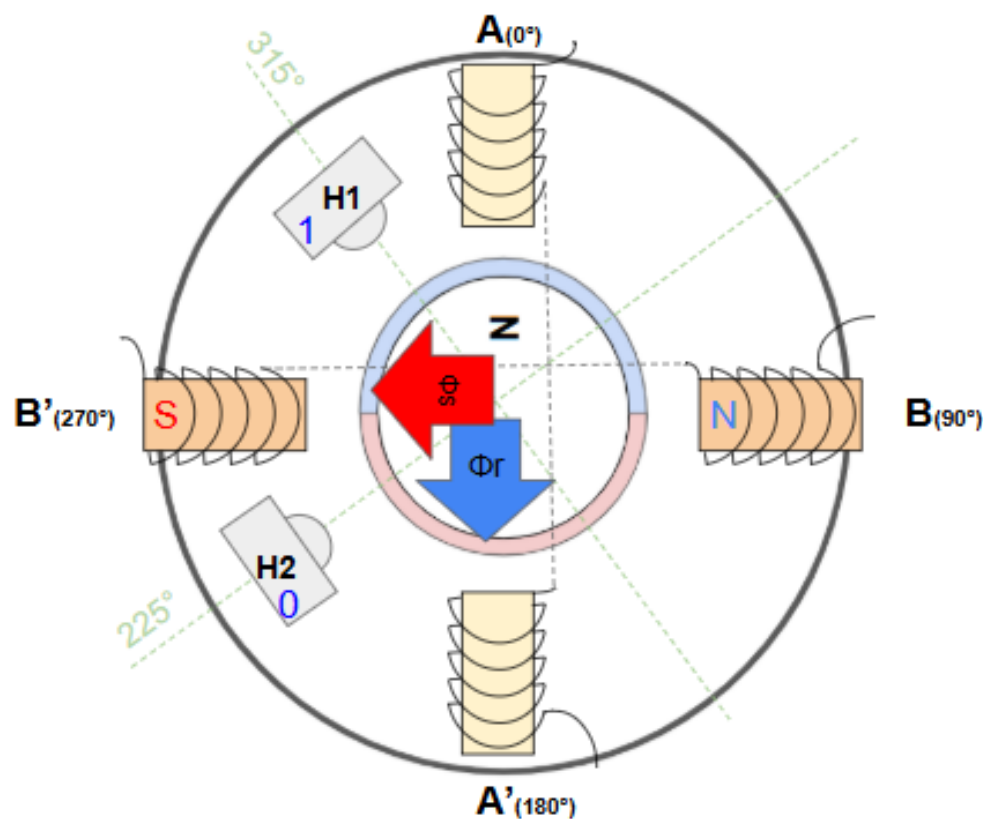


Figure 1. Flux orientation in the machine which is representing step 1

Hall Sensor Design

The concept of my design is to use 2 different coils total, with each representing a positive and negative polarity of the separate excitable phase of operation. A+, A-, B+, B-, these coils will be aligned with the four cardinal directions of the stator

with the positive and negative polarities. This can be done using two Hall sensors, both of which output a high (1) signal when observing a North polarity of magnetic field, and a low (0) signal when observing a South polarity of a magnetic field. These Hall sensors will be placed 90 degrees apart.

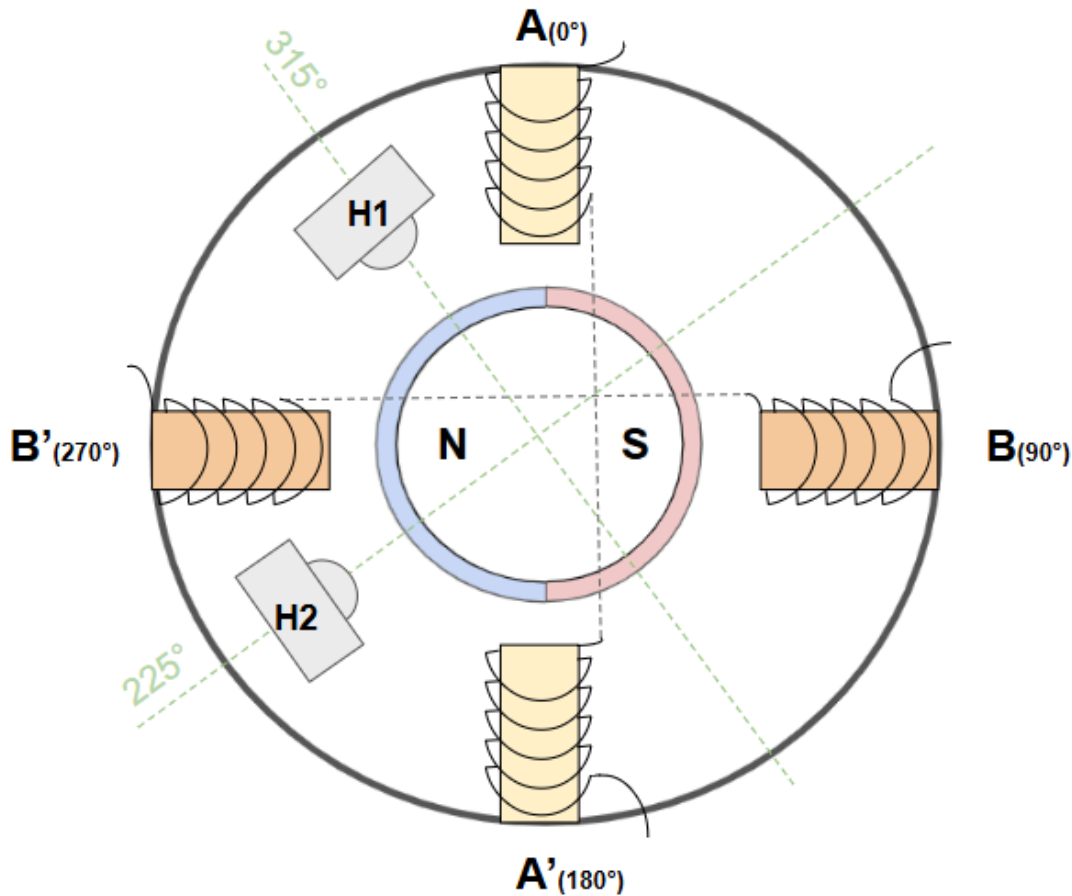


Figure 2. Proposed Hall sensor and coil placement

The placement of the rotor is having two permanent magnets with each one spanning 180°, one with a North polarity and the other with South polarity. This motor design requires four excited coil phases. Starting at the most North (0°) end of the stator the A coil would be placed here. Next, rotating clockwise to East (90°) is where the B coil would be located. Then, rotating again clockwise to South (180°) is where the A' coil would be placed. Finally, rotating clockwise again to

West (270°) the B' coil would be placed. This design for the coils was selected because of the requirement of the design system to have four excited phases during operation. Thus, the need for a power electronic configuration that can provide four different current configurations is required. Considering the use of the Hall sensors two are needed to produce four differentiable states (00), (01), (10), (11). Next, placing the Hall sensors needs to consider how the motor is needed to be operated, the motor should have constant torque on the shaft. Which again returning to how four states are required for operation the placement of the Hall sensors need to account for this. Placing the two Hall sensors 90° apart ensures four distinct, sequential output states per rotation, providing accurate rotor positioning. For my design I will have the Hall sensors placed at 225° and 315° , although they can be anywhere as long as they are 90° apart and not directly on a winding because the winding would produce a magnetic field that would cause the Hall sensors to inaccurately report the rotor position.

System Block Diagram

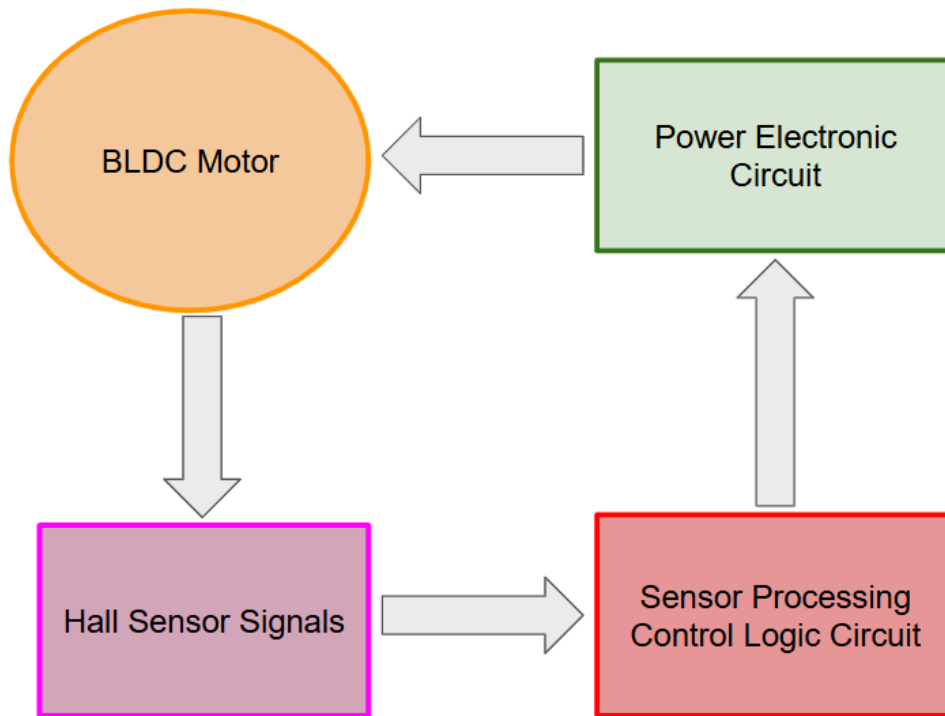


Figure 3. Full System Block Diagram

The system block diagram involves four main stages, which operate in a cyclical pattern. Beginning with the Hall sensor signals block, which is responsible for the signals that the Hall sensors will be outputting. This is transferred to the sensor processing control logic circuit, this is responsible for understanding what the Hall sensors are outputting and providing adequate response to the Hall sensors signals. This is then transferred to the power electronic circuit which takes the processed logic and provides the correct pole to the correct coil in the motor. Then, completing the loop the Hall sensor signals come from the BLDC motor block.

Power Electronics

The power electronics are responsible for providing adequate current control for each of the coils to produce each of the excited states required. This will be done using a H bridge for both the A and B coils to provide a DC motor drive and will provide the coils with negative and positive polarities when required.

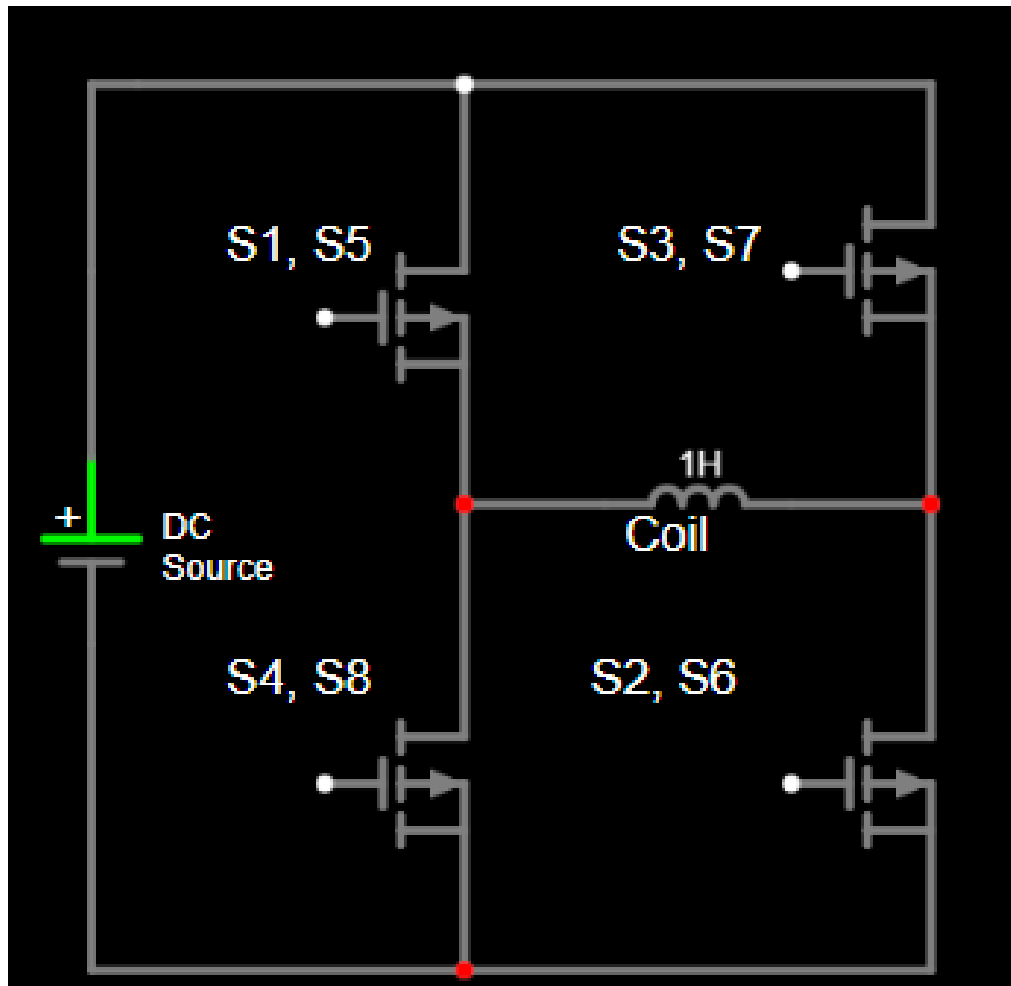


Figure 4. Power electronics circuit design

The figure above is a basic circuit diagram for a coil in the motor, operated by four different MOSFETS (S1-S4). This allows current to flow through the coil in both polarities, which of course will produce the different excited phases for operation. For example for a negative polarity to be generated in the coil S3 and S4 would be provided a high signal allowing current to flow this way. This of course would need

to be constructed for both the A and B coils of the motor, with the respective MOSFETs for each coil with A having S1-S4, and B having S5-S8 .

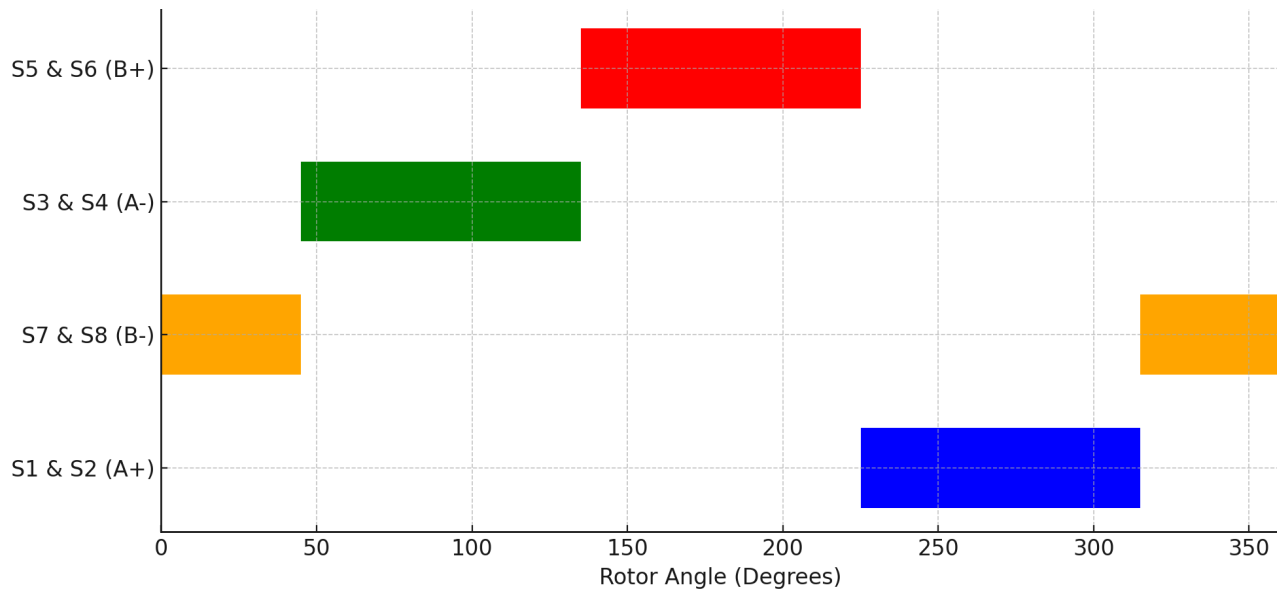


Figure 5. Switch timing diagram

In *Figure 5* each of the different excited phases is expressed in the switches that need to be activated as well as the degrees which they need to be activated. It is important to note that the next phase in the sequence activates when the last phase is half way through its cycle (or 45 degrees into rotation). This allows for the constant torque production in the motor. If this did not occur there would be points in the cycle where there was no pull or push on the rotor magnets.

Digital Logic

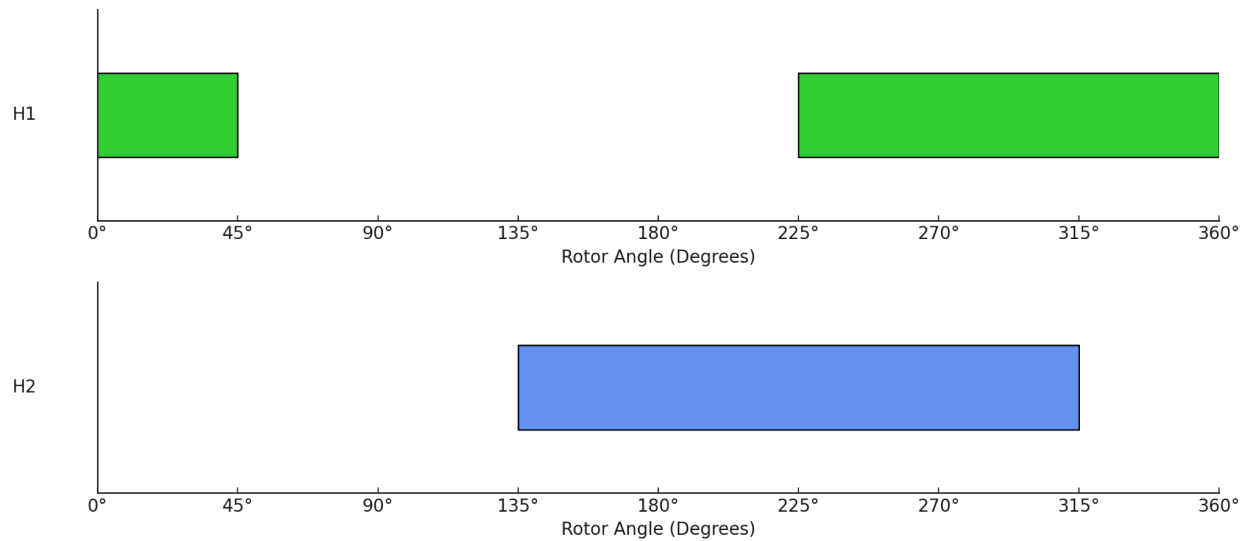


Figure 6. Hall Sensor signal timing diagram

Step	Rotor Position (Degrees)	Switch On
1	(315-45)	(S5 and S6)
2	(45-135)	(S3 and S4)
3	(135-225)	(S7 and S8)
4	(225-315)	(S1 and S2)

Table 1. Step operation of the machine

For *Table 1* rotor position is describing the center point of the North permanent magnet of the rotor.

Truth tables:

Coil to Energize	Step 1	Step 2	Step 3	Step 4
A+	0	0	0	1
B+	1	0	0	0
A-	0	1	0	0
B-	0	0	1	0
Hall Sensor				
H1	1	0	0	1
H2	0	0	1	1

Table 2. Truth table design of the machine

When observing this truth table the relationships that when H2 is low A+ is high, when H1 is low B+ is high, when H2 is high A- is high, and when H1 is high B- is high are seen.

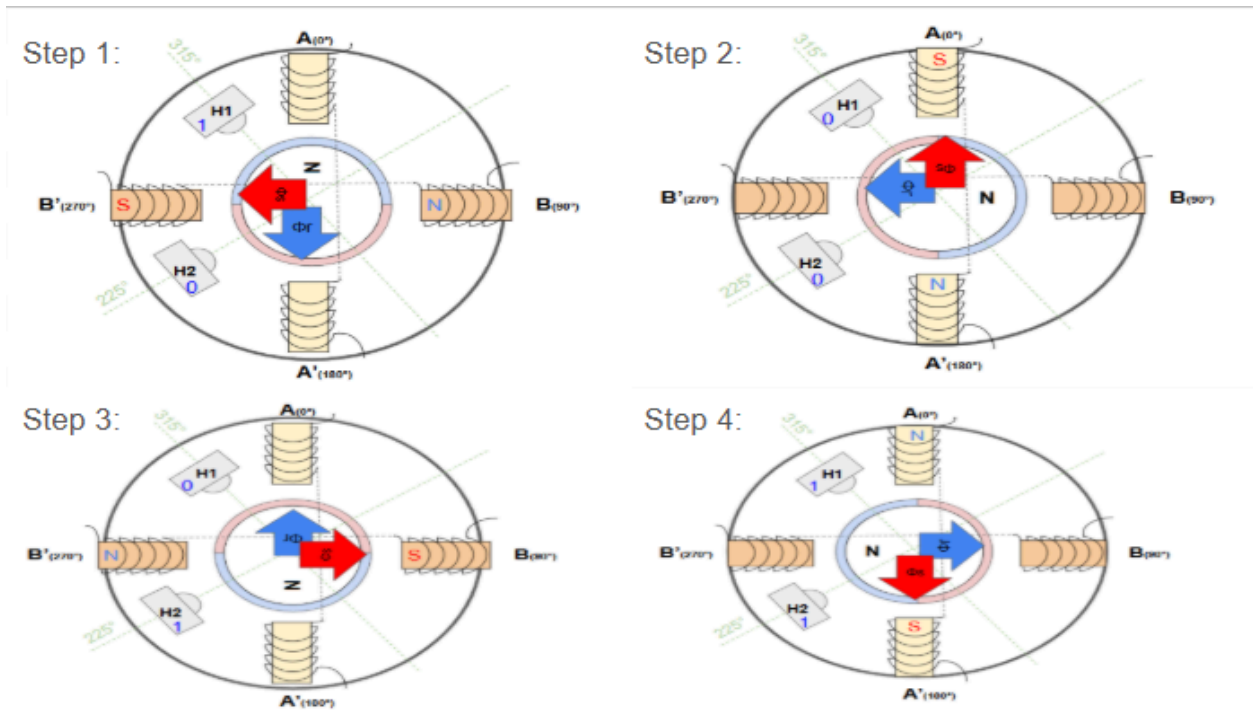


Figure 7: Flux Orientation of each step during full revolution

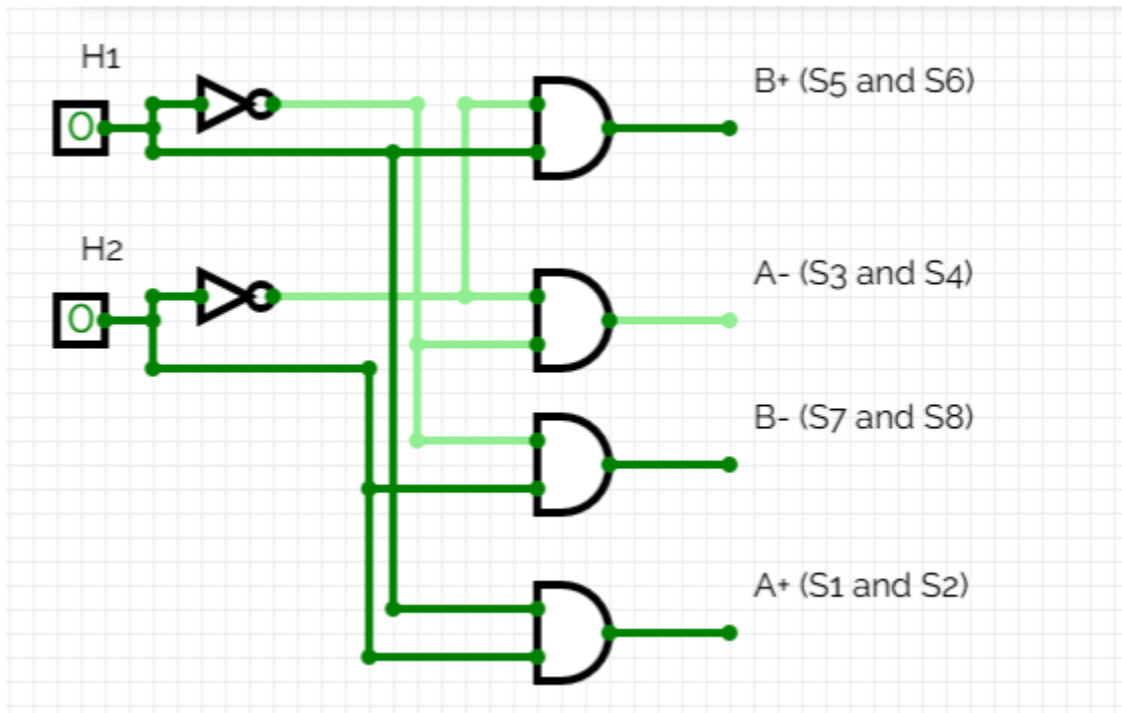


Figure 8. Final logic circuit

The first step in developing this design was to figure out when the Hall sensors would be outputting what. For example when the rotor North is between 45 and 135 degrees, the Hall sensors would both output a zero. This had to be done considering the size of permanent magnets on the rotor both spanning 180 degrees, the final output of this can be seen in *Figure 6*. From this the position of the rotor is known, and next need to figure out what the next position should be and how to get there. Again considering the configuration of the rotor North being located between 45 and 135 degrees the coils to continue clockwise motion on the rotor is what's needed to activate the coil that will be 90 degrees ahead of the current location. Which in this case would be A- configuration, which is turned on with their respective switches as described in *Table 1*. Based on the truth table shown in *Table 2* the relationship between the needed coil to energize and the Hall sensor output can be determined; the flux orientations during a cycle can be seen in *Figure 7*, and the logic constructed into *Figure 8*.

Summary and Conclusion

In this project, a rotor position detection and commutation control system was designed for a two-phase Brushless DC motor using Hall effect sensors. The main requirement for this motor was to generate four switching events (A+, B+, A-, B-) needed for motor drive operation, based on real-time rotor position. Two Hall effect sensors were placed 90° apart around the stator at 225° and 315°, providing information to monitor the rotor's electrical revolution into four sectors. These Hall sensor signals were processed using a logic circuit. The switching sequence was planned to ensure that that clockwise torque was always produced. Two coils were always energized at any given time (as stated in the design problem).

For a 3 phase BLDC motor the minimum investment required inters of magnets for the rotor are 2, very similar to the rotor described in this report both spanning 180 degrees one North and one South. However, for a 3 phase motor this would require three Hall sensors to operate correctly, this is because of the increased amount of phases of three thus more Hall sensors are required to collect more accurately the rotor position. And three sensors would produce more bits of data that can be used and needed for the correct operation of the motor. The geometrical positioning of the Hall sensors should be 120 degrees apart from one

another. And again should not be directly with a stator because the stator will produce a magnetic field that would interfere with the Hall sensor's accuracy of rotor position this can be seen in *Figure 9*.

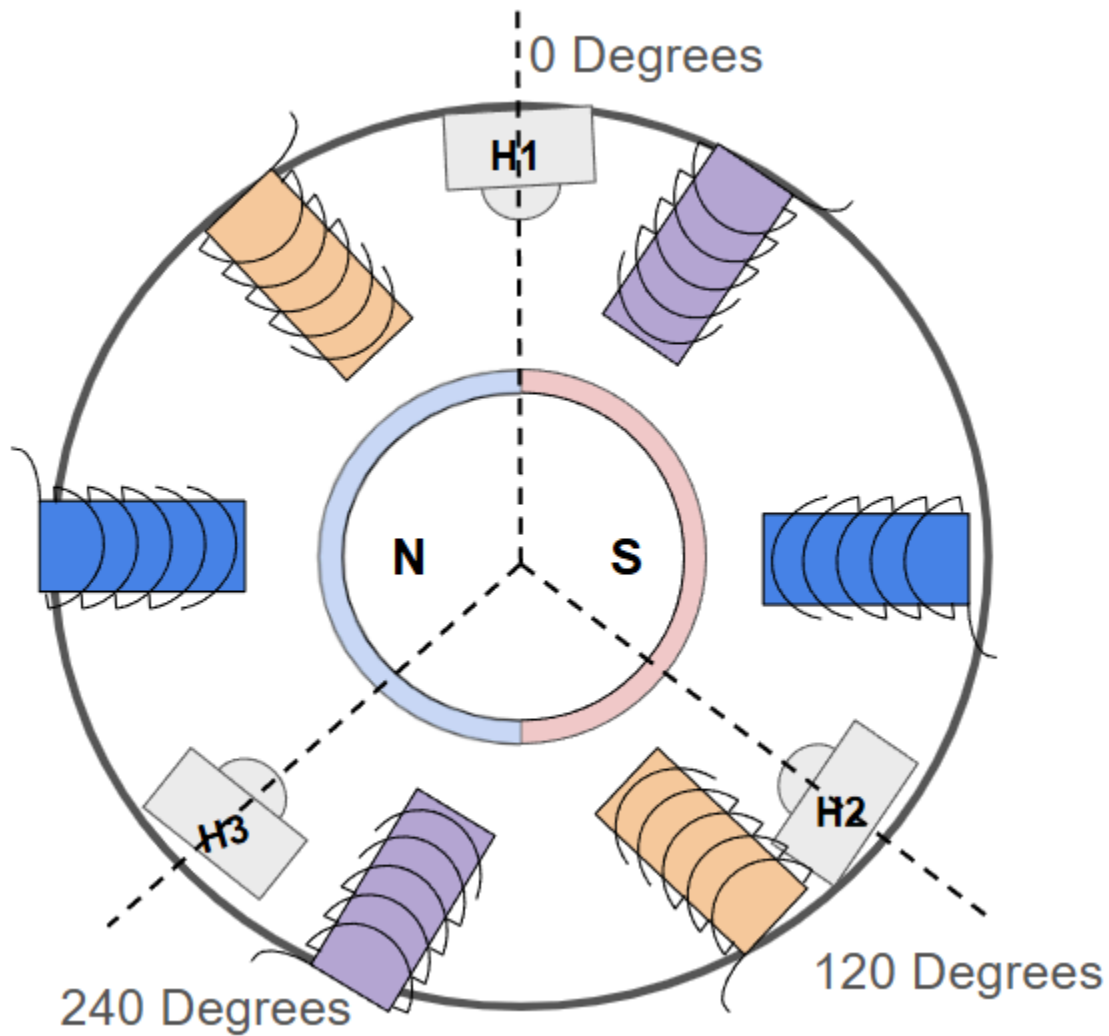


Figure 9: 3 Phase BLDC Motor